Traceable Human–AI Clinical Decision Support: Deterministic Resolution Frontiers for Auditable Reasoning

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***Abstract*—Clinical artificial intelligence can assist or execute predefined reasoning steps, but trustworthy decision support requires knowing whether the representation available to the machine still contains the information needed for a requested operation. This study formalizes an operation-specific criterion for finite deterministic clinical decision support. Each instantiated clinical state is represented as a labeled ternary row in a finite data matrix. A declared transformation chain maps the full labeled state to layer-wise counts, global counts, and a terminal class. Exact support is defined by whether the requested operation can be recovered entirely from a given representation level. The deepest representation level that preserves exact recovery is termed the resolution frontier, and each nonterminal frontier is certified by an explicit recovery rule together with two rule-realizable states that become indistinguishable after the next transformation while requiring different operation outputs. In a 25-parameter representation for infection prophylaxis and vaccination in immunosuppressed adults, four clinically interpretable operations realize the four declared frontier levels. An independent 9-parameter integrity realization reproduces the same profile using fully observed binary inputs. The method therefore identifies when a downstream component may continue an exact operation and when richer information or human review is required. For biomedical research, this provides a directly auditable safeguard for clinical decision support while preserving final clinical authority with the qualified human expert.**

***Index Terms*—clinical decision support, human–AI collaboration, trustworthy artificial intelligence, traceability, formal methods, knowledge representation, biomedical informatics.**

I. Introduction

ARTIFICIAL intelligence is moving from isolated prediction tasks into clinical workflows in which machine outputs are interpreted, transformed, combined with other information, and acted upon by people. Recent healthcare guidance therefore treats traceability, usability, explainability, and human oversight as lifecycle requirements rather than optional presentation features [1]. A 2026 scoping review of 140 empirical human–AI collaboration studies likewise found that benefits depend on task fit, workflow integration, training, and appropriately calibrated trust, while accountability and patient-safety questions remain comparatively under-evaluated [2]. These concerns become sharper when a downstream user receives only an aggregate representation or terminal recommendation rather than the structured state from which it was produced.

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Provenance alone does not close this gap. Knowing where an output originated does not reconstruct why a particular decision was supported, which distinctions were available when it was made, or which distinctions had already been erased by an earlier transformation [3]. In a clinical workflow, this distinction is operational: a terminal class may be correct and fully traceable to its source while still being too coarse to support a later question about a specific vaccine, the location of a risk condition, or the number of nonbinary coordinates. A human reviewer cannot recover information that was not transmitted merely by being the final authority in the workflow.

The present study addresses this narrower but foundational problem. It does not propose another explanation layer for an opaque model and does not claim that every medical AI component should be internally deterministic. Instead, it formalizes a deterministic decision-support layer in which the state presented to a machine, every declared transformation, and the operation requested from the resulting representation are explicit. The central question is: *how far may a clinically structured state be transformed before a specified operation is no longer exactly justified by the representation that remains?*

Recent J-BHI work illustrates two adjacent needs. Clinical explanation research distinguishes faithfulness from usefulness to clinicians [7], while structured decision-tree and knowledge-graph approaches seek explicit medical knowledge representations that can support more reliable reasoning and decision support [8], [9]. The present contribution is complementary to both. It does not score a generated explanation or extract clinical knowledge; it tests whether a requested operation is mathematically determined by the representation that was actually transmitted.

This question complements, but does not duplicate, several existing lines. Clinical guideline formalisms already distinguish conceptual, computable, and implementable representations [4]. Abstract interpretation and property-directed quotients formalize when a property can be preserved under abstraction [5]. Biomedical informatics has recently renewed its call for rigorous theory in knowledge representation, reasoning, and system architecture [6]. Separately, prior formal work has shown that informational substitution does not by itself transfer persistent decision authority [10], and that structural learning can be made reconstructible while preserving a human-defined governing specification [11]. Those are different claims. The contribution here is the missing information-boundary criterion: an exact, operation-specific frontier for the representation actually available to a downstream human–AI process.

The study makes four contributions. First, it states the data object in conventional terms: a finite matrix with one labeled parameter per column and one instantiated state per row. Second, it defines an exact recovery frontier on a declared four-level representation chain. Third, it certifies each nonterminal frontier constructively using rule-realizable clinical states rather than arbitrary symbolic vectors. Fourth, it interprets the frontier as a traceability guardrail: beyond the frontier, an AI component cannot legitimately present the corresponding operation as exactly supported by the transmitted representation; the appropriate action is to obtain richer evidence, expose the limitation, or defer to the qualified human expert.

II. Methods

A. Finite clinical matrix and realizable state space

Let $\Sigma = \{0, 1, \mathrm{U}\}$. Across m instantiations, a finite clinical representation is written as

$$M \in \Sigma^{m\times n}, \tag{1}$$

with one semantically labeled parameter per column and one instantiated state per row. Each row is a labeled vector $v^{(k)} \in \Sigma^n$. Thus a 9-parameter representation has an ambient state space of $3^9 = 19{,}683$ ternary vectors, and a 25-parameter representation has $3^{25} = 847{,}288{,}609{,}443$ formal vectors. Domain rules generally make only a subset realizable. We therefore distinguish the ambient space Σ^n from

$$X = C(D) \subseteq \Sigma^n, \tag{2}$$

where D is the declared domain and C is the deterministic rule map. All separating states used below belong to X; none is created by unconstrained coordinate splicing.

The originating literature calls a fixed parameter semantics together with its admissible ternary state structure a *cell* [12]; conventional matrix/vector terminology is used throughout this paper. The three values are atomic states. In the clinical realization below, U is produced by an observed condition falling inside a declared nonbinary rule region; it is not used as a synonym for missing data, pending work, probability, or confidence. The observation-to-state transduction used by the source rules is explicitly typed and deterministic [13].

The columns may be grouped into semantically declared layers. For the realizations studied here, $n = b^2$ and the n columns are partitioned into b labeled layers $I_1, \dots, I_b$, each containing b parameters. The layer partition is domain structure, not a geometric matrix dimension, and no exchangeability of coordinates within a layer is assumed.

B. Declared representation chain

For $v \in X$ and layer I_j, let $n_{0,j}(v)$, $n_{1,j}(v)$, and $n_{\mathrm{U},j}(v)$ denote the counts of 0, 1, and U in that layer. Let $n_0(v)$, $n_1(v)$, and $n_{\mathrm{U}}(v)$ denote the corresponding global counts. Define four levels $\mathcal{F} = (F_0, F_1, F_2, F_3)$:

$$F_0(v) = v, \tag{3}$$

$$F_1(v) = ((n_{0,1}, n_{1,1}, n_{\mathrm{U},1}), \dots, (n_{0,b}, n_{1,b}, n_{\mathrm{U},b})), \tag{4}$$

$$F_2(v) = (n_0, n_1, n_{\mathrm{U}}), \tag{5}$$

$$F_3(v) = c_D(v), \tag{6}$$

where $c_D : X \to K_D$ is the declared terminal classifier. The terminal codomain K_D is typed separately from Σ. There are deterministic aggregation maps r_j such that $F_{j+1} = r_j \circ F_j$ for $j = 0, 1, 2$.

Table II states exactly what is retained and discarded at each step. The order $0 < 1 < 2 < 3$ is local to this chain; it is not a universal ranking of all representations.

C. Exact support and resolution frontier

Let $Q : X \to Y$ be a specified downstream operation. Exact support from level j means that there exists an explicit recovery map $q_j : F_j(X) \to Y$ such that

$$Q = q_j \circ F_j. \tag{7}$$

Throughout this paper, *resolution* refers to representational resolution; it is not an operation that changes a U coordinate. The operation-specific resolution frontier is

$$\ell(X, \mathcal{F}; Q) = \max\left\{ j \in \{0,1,2,3\} \;\middle|\; \begin{array}{l} Q = q_j \circ F_j \\ \text{for some } q_j \end{array} \right\}. \tag{8}$$

Because $F_{j+1} = r_j \circ F_j$, recoverability from $j+1$ implies recoverability from j. Since F_0 is the identity, the supported levels form a nonempty initial segment.

For $j < 3$, a frontier at j is certified by two ingredients: an explicit q_j, and $x, y \in X$ satisfying

$$F_{j+1}(x) = F_{j+1}(y), \qquad Q(x) \neq Q(y). \tag{9}$$

If Q factored through F_{j+1}, equality of F_{j+1} would force equality of Q, contradicting the witness. This elementary separating-pair argument is the verification device used below; no novelty is claimed for functional dependence itself.

In a human–AI workflow, the operational interpretation is direct. If an AI component receives only $F_k(v)$ and $k > \ell(Q)$, exact execution of Q is not supported by that representation. Greater model capacity cannot recover a distinction that the supplied representation no longer contains. The system must obtain richer information, change the requested operation, or expose the insufficiency to the human decision maker.

Traceability guardrail: More generally, if the downstream interface transmits only $H = \phi \circ F_k$, an exact claim for Q is admissible from that interface only when

$$\exists q \;\; Q = q \circ H. \tag{10}$$

If no such factorization exists on X, no downstream algorithm using H alone can be guaranteed to return Q exactly for all realizable states. Any exact result must then depend on declared side information, a richer representation, or an additional human determination. This is an elementary consequence of functional dependence, used here as an auditable interface rule rather than as a new factorization theorem.

D. Clinical realization: prophylaxis and vaccination

The primary realization uses a previously specified 25-parameter deterministic representation for infection prophylaxis and vaccination in adults with hematologic malignancies and immunosuppression [14]. Its 25 labeled columns are grouped

TABLE I
DECISION-TRACE CHAIN FOR DETERMINISTIC HUMAN–AI CLINICAL SUPPORT. THE FORMAL CONTRIBUTION OF THIS PAPER IS THE REPRESENTATION-SUFFICIENCY TEST IN THE FOURTH ROW; THE REMAINING ROWS DEFINE THE WORKFLOW CONTEXT IN WHICH IT IS USED.

Stage	Machine role	Trace retained	Formal safeguard	Human role
Observation	Capture or structure declared observations	Source observation and admissibility criterion	Typed, declared input rule	Validate domain meaning and clinical applicability
Parameter constitution	Apply predeclared rules	Observation $\rightarrow$ labeled state coordinate	Deterministic state assignment	Define or approve the clinical rule set
State representation	Assemble the row vector	All coordinate identities and ternary values	Full labeled representation F_0	Inspect the complete state when needed
Transformation / query	Aggregate, transmit, or execute Q	Explicit F_j and recovery map q_j	Require $j \leq \ell(Q)$	Judge whether the machine output is adequate for the clinical task
Frontier exceeded	Do not assert exact recovery	Separating-pair certificate showing lost distinction	No exact Q from the transmitted representation	Request richer evidence, use human judgment, or withhold the operation
Final action	Advisory or preauthorized execution only	Complete recorded path available for review	Representation sufficiency does not confer authority	Retain final clinical authority under the governing workflow

TABLE II
DECLARED REPRESENTATION LEVELS.

Level	Retained information	Information removed
F_0	Full labeled vector	None
F_1	Layer identity and ternary counts within each layer	Coordinate identity within a layer
F_2	Global ternary counts	Layer location
F_3	Terminal class only	Exact global counts

into five declared clinical layers: baseline immune status (P01–P05), malignancy/treatment (P06–P10), infectious history (P11–P15), vaccination status (P16–P20), and prophylaxis/follow-up (P21–P25). Current oncology guidance treats influenza, pneumococcal, and other vaccination decisions as distinct clinical objects [15], while infectious-disease guidance separately addresses antimicrobial prophylaxis in cancer-related immunosuppression [16]. The representation is a theoretical and technical demonstrator, not a validated patient-level clinical decision-support system. No cohort, outcome, calibration, or treatment-effect claim is made.

The terminal threshold is $T = 19$. Four operations are evaluated:

- $Q_{\mathrm{vac}}(v) = (v(P16), \ldots, v(P20))$, the labeled vaccination subvector;
- $Q_{\mathrm{loc}}(v)$, the set of declared layers containing at least one 1 or U;
- $Q_{\mathrm{U}}(v) = n_{\mathrm{U}}(v)$, the global count of U coordinates;
- $Q_{\mathrm{class}}(v) = F_3(v)$, the terminal class.

All witness states are established by direct substitution into the published rule regions. The complete 25-coordinate assignments and the cross-coordinate compatibility certificate are supplied in the Supplementary Material.

E. Independent integrity realization

To test whether the frontier construction depends on the clinical semantics or on the use of U, we also use a separately specified 9-parameter AI-system-integrity representation [17]. Its three layers describe model integrity, operational security, and oversight/software-supply-chain controls, with threshold $T = 7$. All witness inputs are fully observed Boolean conditions mapped deterministically to 0 or 1; no U is needed. This realization is not a component of the clinical model and carries no clinical authority claim. It is only an independent structural stress test of the same four-level method.

III. RESULTS

A. Clinical frontiers

Table III summarizes the four clinical results. For Q_{vac}, recovery from F_0 is coordinate projection. A rule-realizable pair places the two nonzero vaccination conditions at different labeled vaccine coordinates while keeping the vaccination-layer counts equal to $(3, 2, 0)$ and all other layers identical. Thus F_1 is identical but Q_{vac} differs, so $\ell(Q_{\mathrm{vac}}) = 0$.

For Q_{loc}, F_1 is sufficient because the layers whose second or third counts are nonzero can be selected directly. One realizable state has P02$= 1$ from an observed CD4 count below 200 cells/μL while active correct PJP prophylaxis keeps P21 at 0; the paired state has P25$= 1$ because no explicit reevaluation plan is present. Both have $F_2 = (24, 1, 0)$, but the affected layers are $\{1\}$ and $\{5\}$. Hence $\ell(Q_{\mathrm{loc}}) = 1$.

For Q_{U}, the third coordinate of F_2 gives exact recovery. A baseline state has $F_2 = (25, 0, 0)$ and terminal class APTO. A second realizable state has only P02 changed to U by an observed CD4 value of 350 cells/μL, lying in the published 200–499 rule region. Its counts are $(24, 0, 1)$ and its terminal class remains APTO because 24 coordinates still satisfy the threshold. Therefore F_3 is identical while Q_{U} differs, proving $\ell(Q_{\mathrm{U}}) = 2$. Finally, $Q_{\mathrm{class}} = F_3$ has frontier 3 by identity.

The selected clinical operation family therefore realizes the profile $(0, 1, 2, 3)$ within the declared chain. This is not asserted as a universal invariant; it is a constructive demonstration that distinct downstream operations on the same clinical state can require different representation depths.

B. Integrity stress test

The independent 9-parameter realization reproduces the same four frontier levels with different semantics. A labeled model-integrity triple has frontier 0; the set of layers containing

TABLE III
OPERATION-SPECIFIC CLINICAL FRONTIERS AND THEIR HUMAN–AI INTERPRETATION.

Operation	Exact recovery	Separating evidence at next level	Frontier	Consequence for a downstream AI component
Labeled vaccination state Q_{vac}	Projection from F_0	Same F_1: vaccination counts $(3, 2, 0)$, but labeled vectors $(1, 0, 0, 0, 1)$ and $(0, 1, 0, 0, 1)$	0	A layer count cannot support a claim about which specific vaccine coordinate differs.
Affected-layer set Q_{loc}	From F_1, select layers with $n_{1,j} + n_{\mathrm{U},j} > 0$	Same $F_2 = (24, 1, 0)$; affected layer $\{1\}$ versus $\{5\}$	1	Global counts cannot support localization of the clinical condition.
Nonbinary count Q_{U}	Third coordinate of F_2	Same $F_3 = \mathrm{APTO}$; $n_{\mathrm{U}} = 0$ versus 1	2	A terminal class cannot establish how many nonbinary coordinates remain underneath it.
Terminal class Q_{class}	Identity on $F_3(X)$	No later declared representation	3	The terminal representation is sufficient only for the terminal-class operation itself.

a compromised control has frontier 1; the number of compromised controls has frontier 2 because states with three and four compromised controls share the same terminal class INDETERMINATE; and the terminal integrity class has frontier 3. All inputs are fully observed and binary. Thus the level-2 result is not an artifact of the U state or of the clinical rules.

IV. DISCUSSION

A. From "explain the answer" to "audit the support"

The practical value of the frontier is not that it makes an opaque model internally interpretable. It changes the object being audited. Instead of asking a model to generate a post hoc reason for its answer, the method records the explicit representation available to the machine, the declared transformation chain, the operation requested, and the exact point at which that operation ceases to be determined by the transmitted state. This distinction is consistent with recent concerns that provenance and generated explanations do not necessarily preserve the operative reason behind a decision [3]. It also differs from evaluating whether an AI explanation is faithful or useful to a clinician [7]: the frontier is evaluated before any explanation is accepted, at the level of whether the requested operation is supported by the information present at all.

The result is therefore stronger than a generic warning that aggregation loses information. If $k > \ell(Q)$, there exist realizable states that are indistinguishable at F_k but require different outputs for Q. No downstream intelligence, human or artificial, can recover the exact operation from F_k alone. A system that nevertheless emits a unique exact answer is importing information from somewhere else, guessing, or concealing an unstated assumption. In a clinical setting, each of those possibilities should be visible to the responsible expert.

This offers a concrete complement to the FUTURE-AI traceability and oversight principles [1]. Logging a model version, data source, and prediction is necessary, but a complete decision trace must also identify whether the representation that reached the decision-support step still contained the distinctions required by the requested operation. Table I makes that chain explicit from observation to final action.

Explicit knowledge structures are already being used in J-BHI to improve clinical decision support, including decision-tree representations and LLM–knowledge-graph integration [8], [9]. The frontier adds a different guarantee to such structured systems: for each downstream operation, it states the latest declared representation from which exact recovery remains possible and provides a witness when the next reduction is too coarse.

B. Human authority and useful machine work

The framework does not require the human to execute every calculation. The machine may apply predeclared rules, assemble vectors, aggregate states, execute exact queries, compare alternatives, and surface a result. What the frontier forbids is presenting an operation as exactly supported after the representation has crossed the boundary at which the needed distinction disappeared.

This information boundary is separate from authority. Prior formal work has shown that even perfect informational substitution does not, by itself, transfer persistent commit authority [10]. The present paper does not reprove that theorem. It uses the separation as an architectural constraint: information sufficiency answers *whether* a machine can support an operation from the evidence it has; clinical governance answers *who* may authorize the consequential action. A highly capable AI can therefore remain useful without becoming the final clinical authority.

Nor does the framework require a static AI system. Traceable structural learning has been formalized separately as a supported, reconstructible change that preserves a human-defined governing specification [11]. Again, that result is not reproved here. Its relevance is that learning and authority need not collapse into one concept: a machine may improve or revise its internal supported knowledge while the clinical workflow continues to require explicit evidence, representation sufficiency, and human authorization at the point of consequential decision.

C. Clinical meaning of the four frontiers

The vaccination witness illustrates why labeled clinical coordinates cannot automatically be treated as exchangeable features. Current cancer-vaccination guidance distinguishes influenza and pneumococcal vaccination as different clinical objects [15]. Two vectors can therefore have identical layer counts but support different vaccine-specific operations. A system receiving only the count has not received the identity.

The localization witness shows a different failure: global counts can preserve severity-like totals while erasing where the condition resides. The U-count witness shows that a terminal class can remain unchanged while a nonbinary coordinate appears underneath it. Importantly, the observed CD4 value is present; the nonbinary state is produced by the declared rule and is not missingness. These examples demonstrate why a terminal recommendation, even when correct for its own purpose, should not be treated as a lossless substitute for the structured state that generated it.

The same principle applies to interfaces. If a downstream component receives only $H = \phi \circ F_j$, any exact recovery claim must factor through H unless additional side information is declared. Information available upstream but not transmitted cannot be silently attributed to the receiver. This is the decision-support analogue of a chain-of-custody requirement: the record must show not only where the output came from, but what information remained available at each transformation step.

D. Limitations

This is a formal methods study, not a clinical-effectiveness study. The 25-parameter representation is a previously published technical demonstrator; no patient cohort, synthetic-patient benchmark, diagnostic accuracy, treatment outcome, calibration, or prospective workflow evaluation is reported. The threshold and parameter rules are inherited rather than revalidated here. The frontier is relative to the declared realizable state space X and to the chosen chain F_0–F_3; a different state space or a different representation chain can produce different frontiers. The integrity realization establishes structural independence from the clinical semantics and from U, not clinical generalizability.

The method also does not claim to expose latent reasoning inside an opaque model. It instead provides a deterministic outer contract: the machine is permitted to make an exact operation claim only when that claim factors through the representation it actually received. Empirical studies are still needed to determine how such contracts affect clinician workload, calibrated trust, error detection, and patient outcomes. The growing human–AI collaboration literature makes those evaluations important [2], but they are downstream of the formal sufficiency result established here.

V. Conclusion

Traceable human–AI clinical reasoning requires more than provenance and more than a terminal answer. It requires a record of the structured state, the transformations applied to it, and the point at which a requested operation is no longer exactly supported by the remaining representation. Operation-specific resolution frontiers provide that boundary. In the clinical realization, four operations require four different levels of the same state, and every nonterminal boundary is certified by a rule-realizable separating pair. The result supplies an auditable guardrail for deterministic decision support: an AI component may perform the work that its available representation supports, but when the frontier is exceeded it must obtain richer information, expose the limitation, or defer. This preserves useful machine assistance while keeping the consequential clinical decision under explicit human authority.

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